

Document Elements

Tables, Lists, Code, Blockquotes & Figures

Tables

1.1 Booktabs Table with Alternating Rows

A clean, booktabs-styled table with dark header and alternating row colors for improved readability in longer data sets.

Table 1. Comparison of Consensus Algorithms

Algorithm	Consensus	Tolerance	Complexity
Paxos	Safety	$f < n/2$	Moderate
Raft	Safety	$f < n/2$	Low
Viewstamped	Safety	$f < n/2$	Moderate
PBFT	Safety	$f < n/3$	High
Zab	Total	$f < n/2$	Moderate
Tendermint	Total	$f < n/3$	Moderate
HotStuff	Total	$f < n/3$	Low

1.2 Multi-Column Table with Merged Cells

Tables can span multiple columns and merge cells for grouped data presentation using the `multirow` package alongside booktabs.

Table 2. Network Protocols by Layer

Layer	Protocol	Transport	Reliable
Transport	HTTP	TCP	Yes
	gRPC	HTTP/2	Yes
	DNS	UDP	No
	TCP	—	Yes
	QUIC	UDP	Yes
Network	IPv4	—	No
	IPv6	—	No

Lists

2.1 Itemize (Unordered)

Custom-styled bullet points with three nesting levels, each using a distinct TikZ-drawn marker for visual hierarchy.

- ▶ Fault tolerance ensures system availability during failures
 - Redundancy is the primary mechanism
 - Active-passive and active-active configurations
 - Geographic distribution across data centers
 - Health monitoring provides early failure detection
 - Automatic failover minimizes downtime
- ▶ Consistency models define data visibility guarantees
- ▶ Scalability enables handling growing workloads
 - Horizontal scaling adds more machines
 - Vertical scaling increases machine capacity

2.2 Enumerate (Ordered)

Numbered lists with styled badges, useful for sequential processes or ranked information presentation.

- 1** Client sends request to the leader node
 - a** Leader appends entry to its local log
 - b** Leader increments log index and term
- 2** Leader replicates entry to follower nodes
- 3** Followers acknowledge successful append
- 4** Leader commits entry once majority acknowledges
- 5** Leader notifies client of successful commit

2.3 Description Lists

Term-definition pairs with bold, colored terms for quick scanning. Ideal for glossaries, feature lists, or configuration options.

Consensus Agreement among distributed nodes on a single data value or system state, despite failures and network delays.

Quorum The minimum number of nodes required to agree before an operation is considered successful, typically a majority.

Leader Election The process by which a distributed system selects a coordinator node to manage replication and coordination.

Heartbeat A periodic message exchanged between nodes to signal liveness and detect failures within bounded time.

Partition Tolerance The ability of a system to continue operating despite network partitions that isolate subsets of nodes.

Code Listings

3.1 Syntax Highlighting

Code blocks with language-aware syntax highlighting, line numbers, and a subtle left accent bar. Built on the `listings` package.

```

1  class RaftNode:
2      def __init__(self, node_id: int, peers: list):
3          self.id = node_id
4          self.peers = peers
5          self.state = "follower"
6          self.term = 0
7          self.log = []
8          self.commit_index = 0
9          self.voted_for = None # None means no vote cast
10
11     def request_vote(self, candidate_term: int):
12         """Handle incoming RequestVote RPC."""
13         if candidate_term > self.term:
14             self.term = candidate_term
15             self.voted_for = None
16             self.state = "follower"
17         return self._grant_vote(candidate_term)
18
19     def append_entries(self, leader_term: int):
20         """Handle incoming AppendEntries (heartbeat)."""
21         if leader_term >= self.term:
22             self.term = leader_term
23             self.state = "follower"
24             return True # Accept entry
25         return False # Reject stale leader

```

Listing 1. Raft-style consensus implementation

3.2 Configuration File Example

Inline code with monospaced font is also styled consistently. Configuration values use `key = value` syntax. The `\texttt{}` command provides inline monospaced text.

```
1 # raft-cluster.toml
2 [cluster]
3 node_id = "node-1"
4 data_dir = "/var/lib/raft"
5 bind_addr = "0.0.0.0:7946"
6
7 [replication]
8 snapshot_interval = "5m"
9 max_log_entries = 65536
10 heartbeat_timeout = "500ms"
11 election_timeout = "1.5s"
12
13 [storage]
14 backend = "boltdb"
15 wal_dir = "/var/lib/raft/wal"
16 snapshot_dir = "/var/lib/raft/snap"
```

Listing 2. Cluster configuration file

Blockquotes

Styled blockquotes with a left accent bar and warm background, suitable for citations, callouts, or important remarks.

“ There are only two hard things in Computer Science: cache invalidation and naming things. — Phil Karlton ”

“ A distributed system is one in which the failure of a computer you didn’t even know existed can render your own computer unusable. — Leslie Lamport ”

“ Any sufficiently complicated C or Fortran program contains an ad hoc, informally-specified, bug-ridden, slow implementation of half of Common Lisp. — Greenspun’s Tenth Rule ”

Figures

5.1 Figure with Caption

A figure environment with a styled caption. Since no image files are available in this demo, we use a TikZ-drawn placeholder.

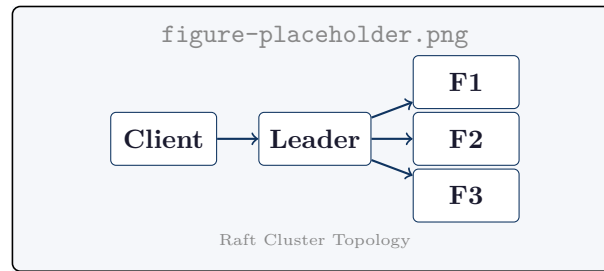


Figure 1. A typical Raft cluster with one leader and three followers handling client requests via log replication.

5.2 Mathematical Content

Mathematical expressions and equations are rendered with consistent styling using the `amsmath` package.

The CAP theorem states that a distributed data store can simultaneously provide at most two of the following three guarantees:

Consistency Every read receives the most recent write or an error

Availability Every request receives a response (not an error)

Partition Tolerance System continues despite network partitions

The probability of quorum agreement in a system of n nodes tolerating f failures is expressed as:

$$Q(n, f) = \binom{n}{f+1} \cdot p^{f+1} \cdot (1-p)^{n-f-1}$$

where p is the probability that any single node responds within the timeout window and f is the maximum number of tolerated failures in a majority-based quorum system.